Fei Yao

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Educational Background

University of Edinburgh, Doctor of Philosophy in Atmospheric and Environmental Sciences	2018.09 – 2022.07
Peking University, Master of Science in Geography (Urban and Regional Planning)	2015.09 – 2018.07
East China Normal University, Bachelor of Science in Geographic Information System	2011.09 – 2015.07

Bio

Assistant Professor (tenure-track) and PhD Surpervisor, China-UK Low Carbon College, Shanghai Jiao Tong University	2026.03 – Present
Research Scientist (open-ended), National Centre for Earth Observation (NCEO) and School of GeoSciences, University of Edinburgh	2023.04 – 2026.02
Postdoctoral Research Associate, School of GeoSciences, University of Edinburgh	2022.03 – 2023.03
Research Consultant (part-time), Edinburgh Innovations, University of Edinburgh	2021.10 – 2022.02
Teaching Assistant (part-time), School of GeoSciences, University of Edinburgh	2019.01 – 2021.09

Research Fields

- Atmospheric environment, its impacts, and responses
- Renewable energy and climate change mitigation
- AI-enabled sustainable energy and environment research

Research Projects

At Shanghai Jiao Tong University:

• New Faculty Start-up Package, China-UK Low Carbon College, Shanghai Jiao Tong University	PI	2026.04 – 2029.04

At the University of Edinburgh:

• Global Change and IT Small Grants, School of GeoSciences, University of Edinburgh	PI	2023.01 – 2024.07
• Google Cloud Research Credits Program and Earth Engine Uplift Program	PI	2022.10 – 2024.01
• IT Small Grants, School of GeoSciences, University of Edinburgh	PI	2021.01 – 2021.07
• Natural Environment Research Council (NERC)	Key Personnel	2023.04 – 2026.02
• UK Space Agency (UKSA)	Participant	2024.01 – 2025.03
• Centre for Earth Observation Instrumentation (CEOI)	Key Personnel	2022.05 – 2023.03

Publications

21 publications in total (11 as first and/or corresponding authors), including 3 in Nature Index journals (as of March 2026).

(Corresponding author*; co-first author#; group members are shown in **bold** when they are first and/or corresponding authors)

21. Liu, J., Zheng, Z., **Yao, F.*** and Li, W.*, 2026. A multi-view machine learning approach for estimating $PM_{2.5}$ concentrations from smartphone photographs. *Journal of Hazardous Materials*, 511, p.142172. doi: 10.1016/j.jhazmat.2026.142172

20. Yin, K.#, **Yao, F.#**, Luo, N., Gao, M., Lu, X.* and Yi, B.*, 2026. Substantial reduction of solar photovoltaic potential in China by an extreme dust event. *Communications Earth & Environment*, 7(1), p.44. doi: 10.1038/s43247-025-03123-1

19. Wang, H., Maslanka, W., Palmer, P.I.*, Wooster, M.J., Wang, H., Yao, F., Feng, L., Wu, K., Lu, X.* and Fan, S.*, 2025. Using geostationary-satellite-derived sub-daily fire radiative power variability versus prescribed diurnal cycles to assess the impact of African fires on tropospheric ozone. *Atmospheric Chemistry and Physics*, 25(23), pp.17501-17526. doi: 10.5194/acp-25-17501-2025

18. **Yao, F.***, Palmer, P.I., Liu, J., Chen, H. and Wang, Y., 2025. Attribution of Solar Energy Yield Gaps due to Transboundary Particulate Matter Pollution Associated with Trade across Northeast Asia. *Environmental Science & Technology*, 59(29), pp.15092-15100. doi: 10.1021/acs.est.5c05935

17. Liu, J.#, **Yao, F.#**, Chen, H.* and Zhao, H.*, 2025. Quantifying the Source–Receptor Relationships of $PM_{2.5}$ Pollution and Associated Health Impacts among China, South Korea, and Japan: A Dual Perspective and an Interdisciplinary Approach. *Environmental Health Perspectives*, 133(3-4), p.047011. doi: 10.1289/EHP14550

16. Wang, Y., Wang, H., **Yao, F.***, Stouffs, R. and Wu, J.*, 2024. An integrated framework for jointly assessing spatiotemporal

dynamics of surface urban heat island intensity and footprint: China, 2003–2020. *Sustainable Cities and Society*, 112, p.105601. doi: 10.1016/j.scs.2024.105601

15. Marvin, M.R.*, Palmer, P.I., Yao, F., Latif, M.T. and Khan, M.F., 2024. Uncertainties from biomass burning aerosols in air quality models obscure public health impacts in Southeast Asia. *Atmospheric Chemistry and Physics*, 24(6), pp.3699-3715. doi: 10.5194/acp-24-3699-2024

14. **Yao, F.*** and Palmer, P.I., 2022. Source Sector Mitigation of Solar Energy Generation Losses Attributable to Particulate Matter Pollution. *Environmental Science & Technology*, 56(12), pp.8619-8628. doi: 10.1021/acs.est.2c01175

13. Liu, J.*, Li, J. and Yao, F., 2022. Source-receptor relationship of transboundary particulate matter pollution between China, South Korea and Japan: Approaches, current understanding and limitations. *Critical Reviews in Environmental Science and Technology*, 52(21), pp.3896-3920. doi: 10.1080/10643389.2021.1964308

12. Wu, J.*, Wang, Y., Liang, J. and Yao, F., 2021. Exploring common factors influencing $PM_{2.5}$ and O_3 concentrations in the Pearl River Delta: Tradeoffs and synergies. *Environmental Pollution*, 285(117138), pp.10-1016. doi: 10.1016/j.envpol.2021.117138

11. Mogno, C.*, Palmer, P.I., Knote, C., Yao, F. and Wallington, T.J., 2021. Seasonal distribution and drivers of surface fine particulate matter and organic aerosol over the Indo-Gangetic Plain. *Atmospheric Chemistry and Physics*, 21(14), pp.10881-10909. doi: 10.5194/acp-21-10881-2021

10. **Yao, F.*** and Palmer, P.I., 2021. A model framework to reduce bias in ground-level $PM_{2.5}$ concentrations inferred from satellite-retrieved AOD. *Atmospheric Environment*, 248, p.118217. doi: 10.1016/j.atmosenv.2021.118217

9. Guo, H., Zhan, Q., Ho, H.C., Yao, F., Zhou, X., Wu, J. and Li, W.*, 2020. Coupling mobile phone data with machine learning: How misclassification errors in ambient $PM_{2.5}$ exposure estimates are produced?. *Science of The Total Environment*, 745, p.141034. doi: 10.1016/j.scitotenv.2020.141034

8. Guo, H., Li, W.*, Yao, F., Wu, J., Zhou, X., Yue, Y. and Yeh, A.G., 2020. Who are more exposed to $PM_{2.5}$ pollution: A mobile phone data approach. *Environment International*, 143, p.105821. doi: 10.1016/j.envint.2020.105821

7. Wu, J., Liang, J., Zhou, L., Yao, F. and Peng, J.*, 2019. Impacts of AOD correction and spatial scale on the correlation between high-resolution AOD from Gaofen-1 satellite and In Situ $PM_{2.5}$ measurements in Shenzhen City, China. *Remote Sensing*, 11(19), p.2223. doi: 10.3390/rs11192223

6. **Yao, F.**, Wu, J.*, Li, W.* and Peng, J., 2019. A spatially structured adaptive two-stage model for retrieving ground-level $PM_{2.5}$ concentrations from VIIRS AOD in China. *ISPRS Journal of Photogrammetry and Remote Sensing*, 151, pp.263-276. doi: 10.1016/j.isprsjprs.2019.03.011

5. **Yao, F.**, Wu, J.*, Li, W. and Peng, J., 2019. Estimating daily $PM_{2.5}$ concentrations in Beijing using 750-M VIIRS IP AOD retrievals and a nested spatiotemporal statistical model. *Remote Sensing*, 11(7), p.841. doi: 10.3390/rs11070841

4. **Yao, F.**, Si, M., Li, W.* and Wu, J.*, 2018. A multidimensional comparison between MODIS and VIIRS AOD in estimating ground-level $PM_{2.5}$ concentrations over a heavily polluted region in China. *Science of the Total Environment*, 618, pp.819-828. doi: 10.1016/j.scitotenv.2017.08.209

3. Wang, Z., Yao, F., Li, W. and Wu, J.*, 2017. Saturation correction for nighttime lights data based on the relative NDVI. *Remote Sensing*, 9(7), p.759. doi: 10.3390/rs9070759

2. Wu, J., Yao, F., Li, W.* and Si, M., 2016. VIIRS-based remote sensing estimation of ground-level $PM_{2.5}$ concentrations in Beijing–Tianjin–Hebei: A spatiotemporal statistical model. *Remote Sensing of Environment*, 184, pp.316-328. doi: 10.1016/j.rse.2016.07.015

1. **Yao, F.**, Ye, K. and Zhou, J.*, 2015. Automatic image classification and retrieval by analyzing plant leaf features. *Journal of Zhejiang A&F University*, 32(3), pp.426-433. doi: 10.11833/j.issn.2095-0756.2015.03.015

Teaching

Teaching Assistant at the University of Edinburgh:

- Earth's Atmospheric Composition (2020, 2021)
- Visual Analytics (2019)
- Welcome Week Computing Induction (2019)

Other Professional Activities

Youth Editorial Board of *Carbon Footprints*; Referee for top-tier journal such as Environmental *Science & Technology*.

Honors and Awards

Student Years (Selected):

- National Scholarship (2013, 2016, 2017)
- Excellent Graduate of Peking University (2018)

- Peking University Award for Academic Innovation (2016, 2017)
- 2016 IACP (International Association for China Planning) Best Student Paper
- Excellent Graduate of Shanghai Municipal (2015)